

Visualizing DARTS CTD Data – JMB Live Coding

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Load packages that we will need to use for plotting data using R. The package is called “tidyverse” which is a huge set of tools that are great for data analysis and visualization.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Now let's load our data frame into R. (the csv we just made)

```
data <- read.csv("240905_darts_clean.csv")
```

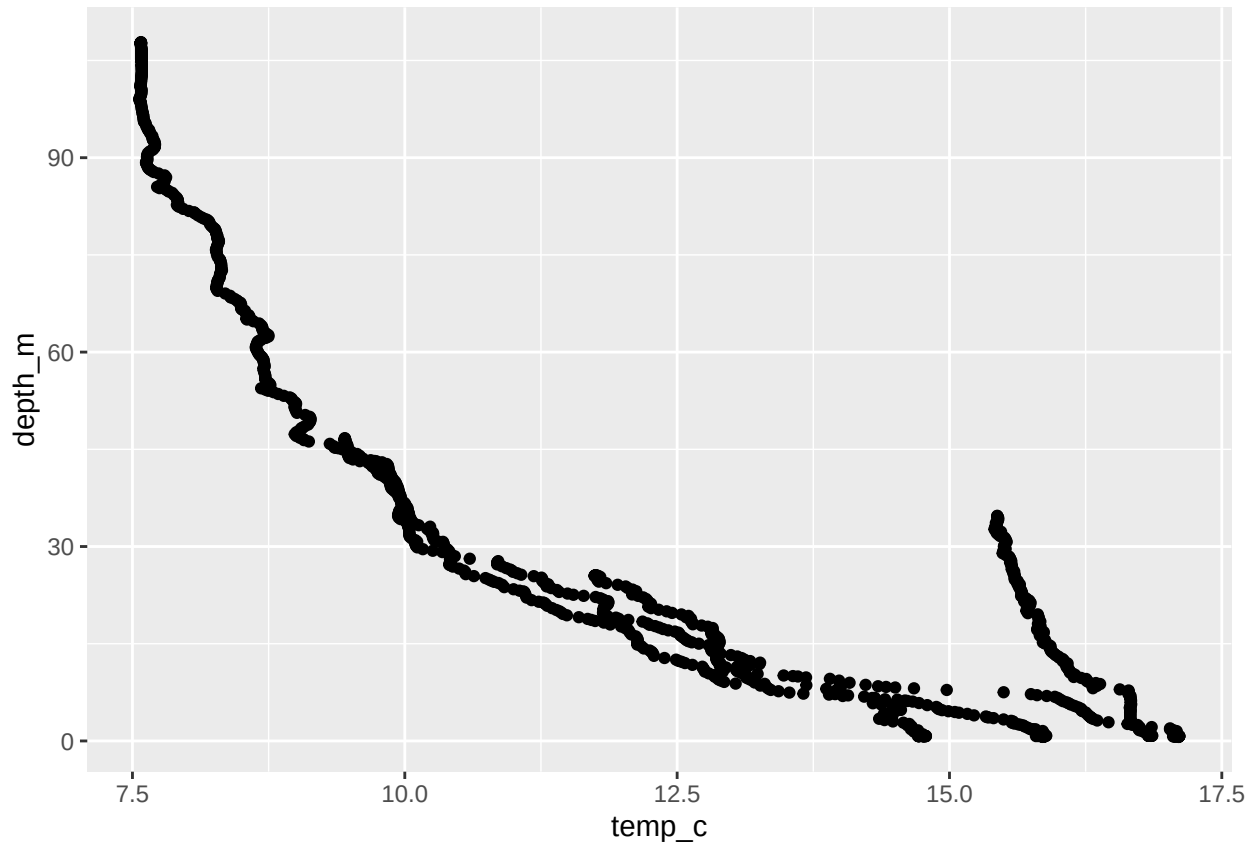
What does our data look like? To see the structure of your data:

```
str(data)
```

```
## 'data.frame':    1258 obs. of  12 variables:
## $ date           : int  20240905 20240905 20240905 20240905 20240905 20240905 20240905 20240905 20240905 20240905 20240905 20240905
## $ station        : chr  "S1" "S1" "S1" "S1" ...
## $ depth_m        : num  0.78 0.787 0.789 0.79 0.792 0.791 0.79 0.791 0.796 0.801 ...
## $ temp_c         : num  16.8 16.8 16.8 16.8 16.8 ...
## $ sal_psu        : num  31.3 31.3 31.3 31.3 31.3 ...
## $ density_kg_per_m3: num  22.7 22.7 22.7 22.7 22.7 ...
## $ par            : num  1400 1700 1670 1790 1640 1660 1490 1560 1430 1550 ...
## $ fluor_mg_per_m3 : num  0.978 0.984 1.007 1.011 1.052 ...
## $ turbidity_ntu   : num  1.16 1.15 1.15 1.15 1.15 ...
## $ O2_umol_per_kg  : num  200 201 201 201 202 ...
## $ O2_pct_sat      : num  81.7 81.8 81.8 81.8 82.3 ...
## $ ph              : num  8.02 8.03 8.03 8.03 8.03 ...
```

Plot something! temperature and depth:

```
ggplot(data, aes(x=temp_c, y=depth_m)) + geom_point()
```

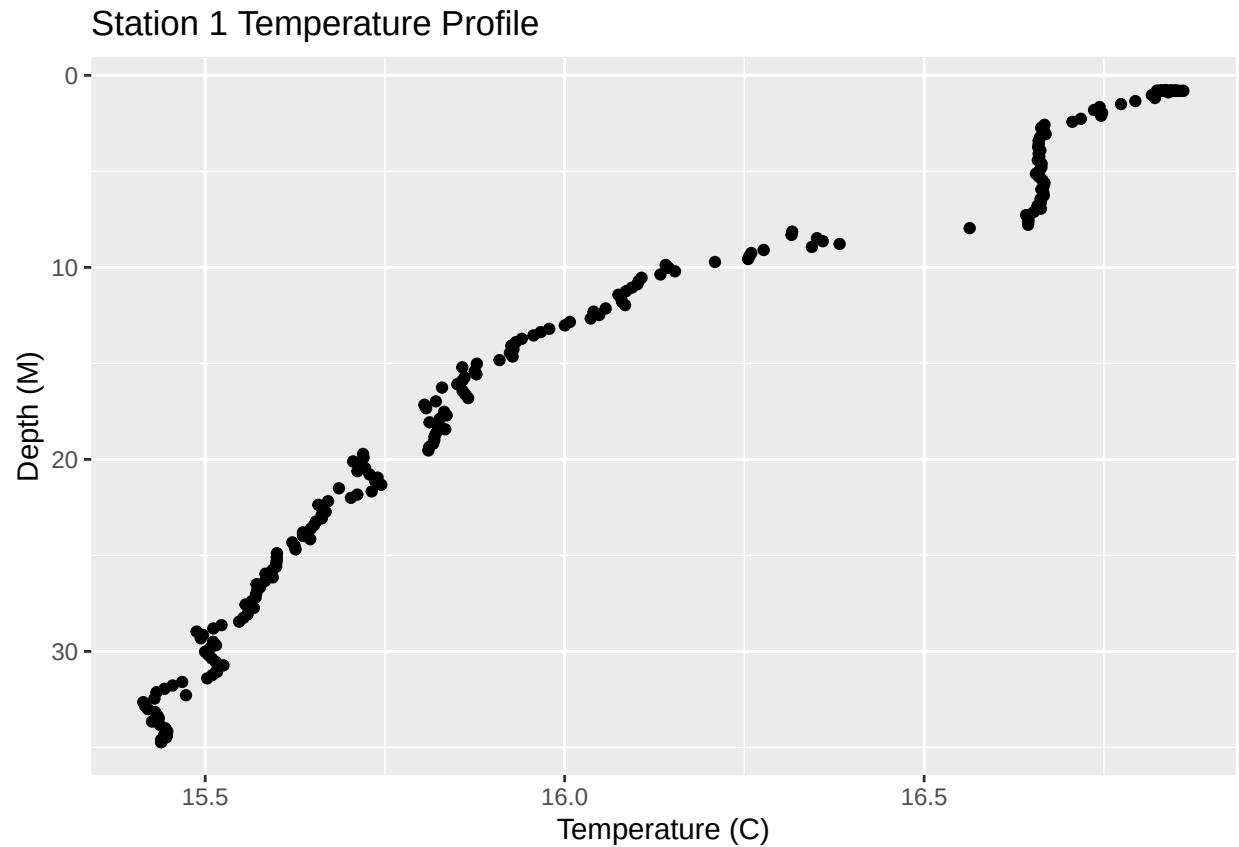


plotting water column temperature profile for station 1:

```
stn1 <- filter(data, station == "S1")
```

== symbol tests for a value to be equivalent to what you provide != choose values not equal to what you provide > “greater than” can apply this test when the column values are integers or numeric values < “less than”

```
ggplot(stn1, aes(x = temp_c, y = depth_m)) +  
  geom_point() +  
  scale_y_reverse() +  
  xlab('Temperature (C)') +  
  ylab('Depth (M)') +  
  ggtitle('Station 1 Temperature Profile')
```



```
ggsave('stn1_temp_profile.png', width=4, height=6)
```

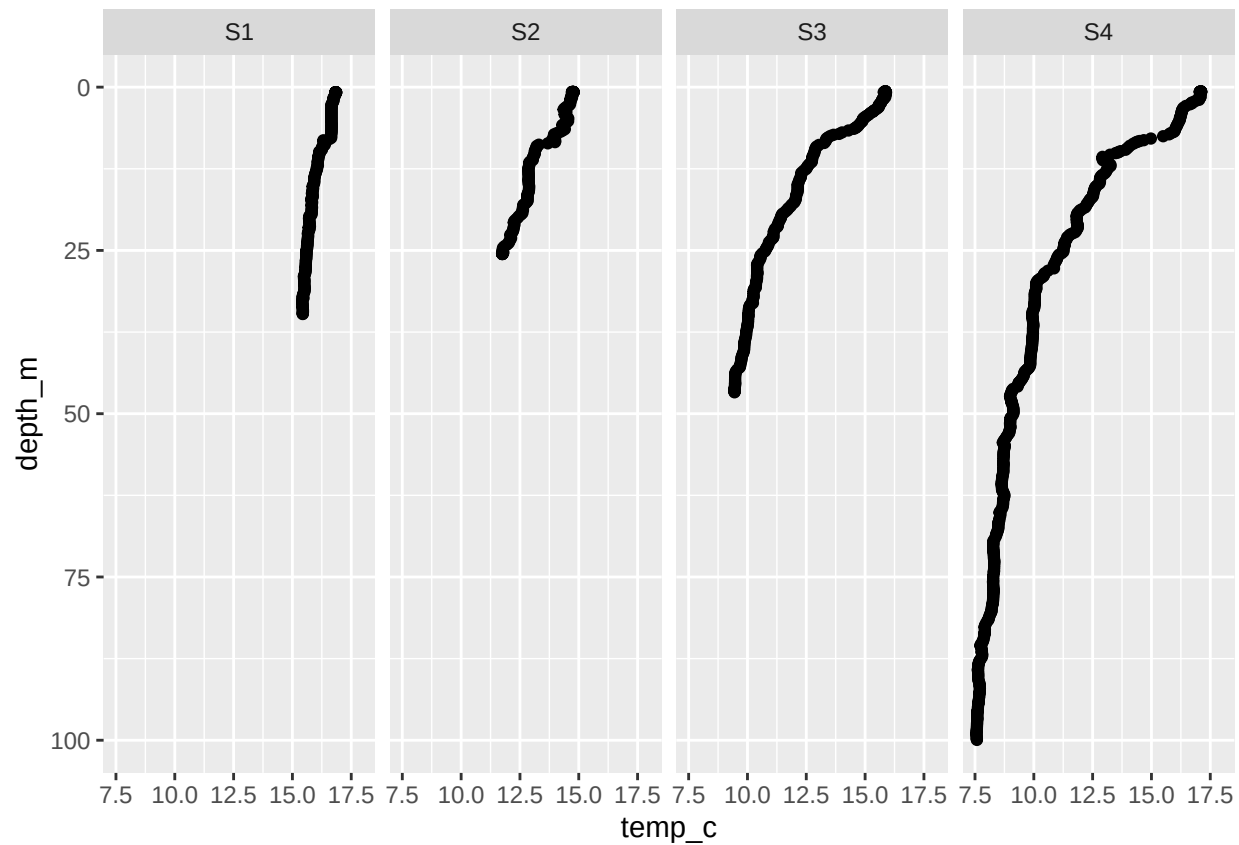
plotting profiles from multiple stations at the same time, in the same figure.

```
ggplot(data, aes(x = temp_c, y = depth_m)) +  
  geom_point() +  
  facet_wrap(~station, ncol = 4) +  
  scale_y_reverse() +  
  ylim(100, 0) +  
  xlim(7.5, 18)
```

```
## Scale for y is already present.
```

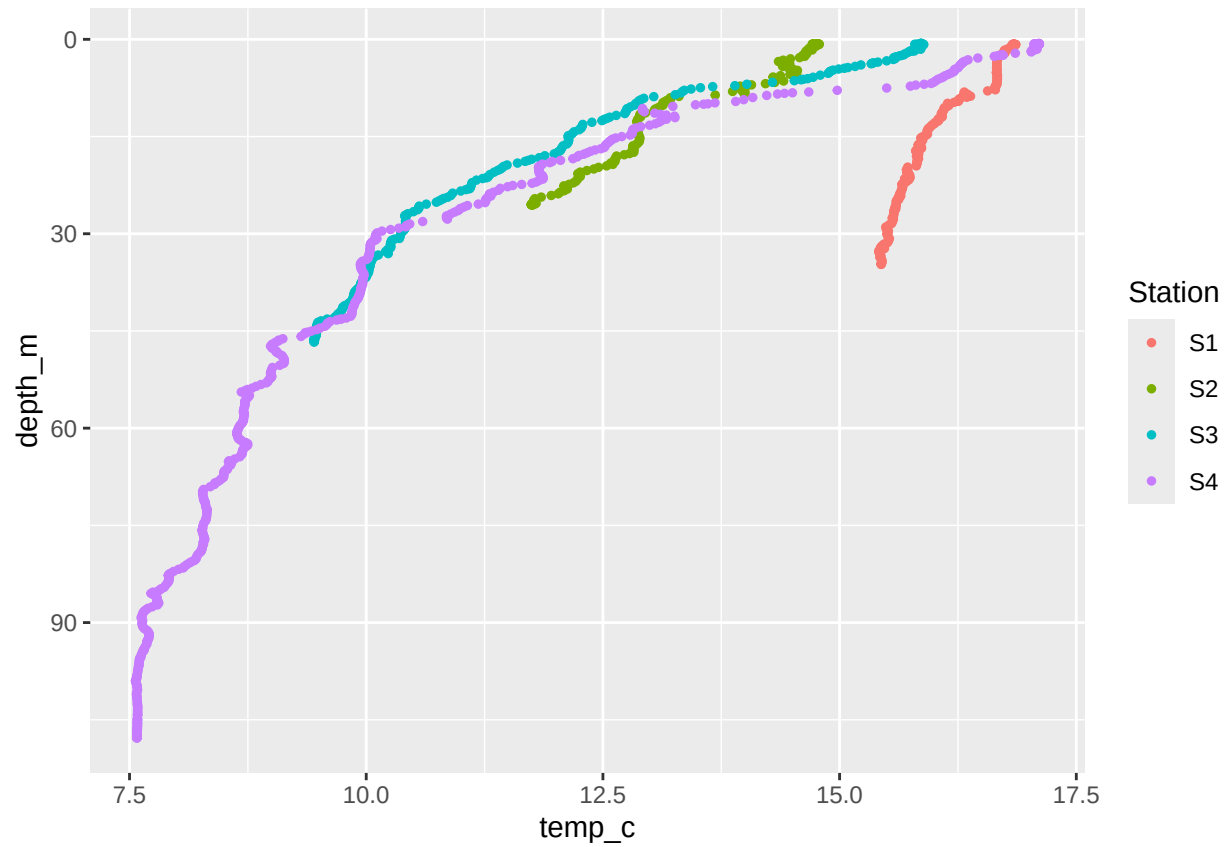
```
## Adding another scale for y, which will replace the existing scale.
```

```
## Warning: Removed 81 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



Plotting all profiles on the same plot:

```
ggplot(data, aes(x = temp_c, y = depth_m, color = station)) +  
  geom_point(size = 1) +  
  scale_y_reverse() +  
  labs(color = 'Station')
```



JMB Scratch after class to play with plotting T-S diagrams for watermasses worksheet:

```
ggplot(data, aes(y = temp_c, x = sal_psu, color = station)) +  
  geom_point(size = 1) +  
  scale_y_reverse() +  
  labs(color = 'Station')
```

